Clinical Study Summary Report (CSSR) Document Status: Final | **Version:** 1.0 | **Date:** 04-Apr-2026

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1. Administrative & Study Identification

Full Study Title: Study of Dose Confirmation and Safety of Crizanlizumab in Pediatric Sickle Cell Disease Patients **Official Title:** A Phase 2, Multicenter, Open-Label Study to Assess Appropriate Dosing and to Evaluate Safety of Crizanlizumab, With or Without Hydroxyurea/Hydroxycarbamide, in Sequential, Descending Age Groups of Pediatric Sickle Cell Disease Patients With Vaso-Occlusive Crisis **Short Title/Acronym:** SOLACE-kids (CSEG101B2201) **Protocol Number:** CSEG101B2201 **NCT Number:** NCT03474965 **Posting ID:** 739628149306423520 **Trial Status:** COMPLETED **Sponsor / Company:** Novartis Pharma Services Inc. / Novartis **Investigational Product:** Crizanlizumab (Trade Name: Adakveo, ATC Code: B06AX01) **Phase of Development:** Phase II **Medical Monitor:** Novartis Clinical Operations Lead / Study Steering Committee

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2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To confirm and establish appropriate dosing (Part A) and to evaluate the safety and efficacy (Part B) of crizanlizumab in pediatric patients ages 2 to <18 years with a history of Vaso-Occlusive Crises (VOC) with or without Hydroxyurea/Hydroxycarbamide (HU/HC). **Secondary Objectives:** To assess the long-term safety, pharmacokinetics (PK), pharmacodynamics (PD), and the reduction in the annualized rate of VOCs over a 2-year period.

2.2 Background & Rationale To address the gap in preventative treatments for pediatric patients suffering from Sickle Cell Disease (SCD) / Anemia, Sickle Cell. Vaso-occlusive crises (VOCs) are the hallmark of SCD, largely driven by P-selectin-mediated cell adhesion. The efficacy and safety of crizanlizumab (a P-selectin inhibitor) have been previously established in adults. This study utilizes PK/PD extrapolation from the adult population to determine safe and effective dosing in pediatric cohorts to mitigate the frequency of debilitating VOCs.

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3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional **Control Type:** Single-arm **Blinding:** Open-label **Randomization:** N/A (Participants are stratified by sequential, descending age groups: Group 1 [12–<18 years], Group 2 [6–<12 years], Group 3 [2–<6 years])

3.2 Planned Enrollment & Duration Targeted Enrollment: 117 evaluable pediatric patients **Number of Sites:** Multicenter (Global) **Estimated Study Start:** 2018 (Historical Initiation) **Estimated Primary Completion:** Up to 2 years per patient

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4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Male or female patients aged 2 to <18 years. 2. Confirmed diagnosis of Sickle Cell Disease (SCD) (any genotype including HbSS, HbSC, HbSβ0-thalassemia, HbSβ+-thalassemia, etc.) by hemoglobin electrophoresis or/and HPLC. 3. Experienced at least 1 VOC within the preceding 12 months, as determined by medical history. The VOC must have resolved at least 7 days prior to first dose, resulted in a medical/healthcare professional visit, and required oral/parenteral opioids or parenteral NSAIDs. 4. If receiving HU/HC, L-glutamine, or an erythropoietin-stimulating agent, must have been on a stable dose for at least 6 months prior to screening. 5. Received standard age-appropriate care for SCD (penicillin prophylaxis, pneumococcal immunization). 6. Performance status: Karnofsky $\ge$ 50% for patients >10 years of age, and Lansky $\ge$ 50 for patients $\le$ 10 years of age. 7. Laboratory requirements: Absolute Neutrophil Count $\ge 1.0 \times 10^9$/L, Platelets $\ge 75 \times 10^9$/L, Hemoglobin (Hgb) > 5.5 g/dL. 8. Renal and hepatic function: eGFR ≥ 75 mL/min/1.73 m^2, Direct bilirubin $\le 2.0 \times$ ULN, ALT $\le 3.0 \times$ ULN. 9. Transcranial Doppler (TCD) for patients aged 2 to <16 years indicating low risk for stroke.

4.2 Exclusion Criteria 1. History of stem cell transplant or gene therapy. 2. Any documented history of a clinical stroke, intracranial hemorrhage, or an uninvestigated neurologic finding within the past 12 months. 3. Received blood products within 30 days prior to Week 1 Day 1 dosing, or planning to participate in a chronic transfusion program. 4. Received a monoclonal antibody or immunoglobulin-based therapy within 6 months of screening. 5. Pregnant females, females who have given birth within the past 90 days, or who are breastfeeding. 6. Patients having taken voxelotor less than 30 days prior to Screening, or planning to take voxelotor while on study. 7. Active HIV infection, active Hepatitis B infection, or known Hepatitis C history. 8. Cardiac or cardiac repolarization abnormalities (e.g., Resting QTcF ≥ 450 msec for <12 years; ≥ 450 msec for males and ≥ 460 msec for females 12 years and older).

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5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: Crizanlizumab administered every 4 weeks with a loading dose 2 weeks after the first dose (i.e., on Week 1 Day 1, Week 3 Day 1, and then Day 1 of every fourth week). *Dosing:* Initial dose of 5 mg/kg for Group

1 and Group 2. Group 2 and 3 dosing was subsequently adjusted to 8.5 mg/kg based on emerging PK data and dose confirmation analysis determined in Part A. **Route of Administration:** Intravenous (IV) infusion. **Duration of Treatment:** Up to 2 years (104 weeks).

5.2 Concomitant & Prohibited Therapy Permitted: Hydroxyurea/ Hydroxycarbamide (HU/HC), L-glutamine, standard age-appropriate care for SCD, and episodic transfusions in response to worsened anemia or VOC. **Prohibited:** Initiation of a chronic transfusion program, monoclonal antibody or immunoglobulin-based therapy within 6 months of screening, use of therapeutic anticoagulation within 10 days prior to first dose, or use of voxelotor.

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6. Study Timeline & Visit Schedule

| Visit ID | Window (Days) | Key Activities/Procedures | | :--- | :--- | :--- | | **Screening** | Day -30 to 0 | Informed Consent, Medical/VOC History, SCD Genotype confirmation, TCD, Labs | | **Baseline** | Week 1, Day 1 | First IV Infusion (Loading Dose 1), PK/PD sampling, Safety Assessment | | **Loading 2** | Week 3, Day 1 | Second IV Infusion (Loading Dose 2), Safety Assessment | | **Interim** | Every 4 Weeks | Maintenance IV Infusions, Safety Labs, Efficacy Assessment (VOC Tracking) | | **End of Study** | Year 2 (Week 104) | Final Vital Signs, Exit Interview, IP Accountability, Final Safety Review |

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7. Outcome Measures (Endpoints)

7.1 Primary Endpoints (Part A) 1. **Pharmacokinetics (PK) - AUCd15:** Area under the curve from time zero to the last measurable concentration sampling time following the first dose of crizanlizumab. 2. **Pharmacokinetics (PK) - AUCtau:** Area under the curve calculated to the end of a dosing interval at steady-state after multiple doses. 3. **Pharmacokinetics (PK) - Cmax:** Maximum observed serum drug concentration after single (first dose) and multiple doses (steady-state).

7.2 Secondary Endpoints 1. Annualized Rate of Vaso-Occlusive Crisis (VOC) Events Leading to Healthcare Visit in Clinic / Emergency Room (ER) / Hospital. 2. Annualized Rate of Vaso-Occlusive Crisis (VOC) Events Treated at Home (Based on Documentation by Health Care Provider Following Phone Contact). 3. Annualized Rate of Vaso-Occlusive Crisis (VOC) Events - Uncomplicated Pain Crisis.

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8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Continuous collection via patient/ caregiver interviews, monitoring during and after IV infusions for infusion-related reactions, and periodic physical exams. **Serious Adverse Events (SAEs):** Standard 24-hour reporting timeline for any event resulting in death, life-threatening conditions, or hospitalization. **Data Monitoring Committee (DMC):** Independent Study Data Safety Monitoring Board providing oversight of unblinded safety and PK/PD data.

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9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Intent-to-Treat (ITT) / Enrolled Safety Set for PK/PD modeling and safety outcomes. **Sample Size Justification:** $N = 117$ powered to detect appropriate therapeutic exposure levels across varying pediatric age/weight strata and to evaluate long-term safety. **Handling of Missing Data:** Last Observation Carried Forward (LOCF) for partial exposure periods when calculating annualized VOC incidence rates.

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10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki. **Monitoring Plan:** Regular onsite monitoring visits combined with centralized evaluation of PK/PD laboratories. **Audit Trail:** eCRF locking, automated alerts for dosing deviations, and data clarification forms for centralized database review.

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11. Study Identifiers & Governance

Primary ID: CSEG101B2201 **Registry Number:** NCT03474965 **Posting ID:** 739628149306423520 **Sponsor/Lead Organization:** Novartis Pharma Services Inc. / Novartis **Study Title:** SOLACE-kids **Official Regulatory Title:** A Phase 2, Multicenter, Open-Label Study to Assess Appropriate Dosing and to Evaluate Safety of Crizanlizumab, With or Without Hydroxyurea/Hydroxycarbamide, in Sequential, Descending Age Groups of Pediatric Sickle Cell Disease Patients With Vaso-Occlusive Crisis

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12. Clinical Framework (Sickle Cell Disease Specific)

Primary Condition: Sickle Cell Disease (SCD) / Anemia, Sickle Cell **Diagnosis Threshold:** History of $\ge$ 1 Vaso-Occlusive Crisis (VOC) in the preceding 12 months **Medical Methodology:** P-selectin inhibition (monoclonal antibody therapy) with/without Hydroxyurea **Optimization Target:** Prevention and reduction of annualized VOCs

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13. Study Procedures & Methodology

Management Pathway: Prophylactic maintenance therapy via recurring IV infusions **Education Protocol (HCP):** Infusion protocol training, PK/PD sampling logistics, and safety monitoring **Education Protocol (Patient):** Age-appropriate SCD care counseling, hydration protocols, and detailed VOC reporting criteria for caregivers **Quality Audit Mechanism:** Central PK data tracking, dosing confirmation reviews (Part A to Part B transition), and automated clinical event logging

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14. Observation & Follow-Up Schedule

Total Duration: 104 Follow-up weeks (2 years) **Onsite Visit Cadence:** Week 1, Week 3, and then Day 1 of every 4-week cycle (Q4W) **Remote Monitoring Frequency:** Intermittent monitoring between infusion cycles for safety and VOC occurrences **Checklist Review Interval:** Prior to every infusion (Q4W) to evaluate concomitant meds, VOC history, and AE presentation

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15. Sign-off & Approval

Role	Name	Signature	Date
Principal Investigator	[Name]	____	[DD-MMM-YYYY]
Clinical Operations Lead	[Name]	___	[DD-MMM-YYYY]
Lead Statistician	[Name]	_____	[DD-MMM-YYYY]